

Customer Segmentation & Marketing Analytics

Executive Summary — eCommerce Platform 2020–2026

Overview

This analysis examined 8,000 customers and 25,000 orders from an eCommerce platform spanning 2020 to 2026. Using RFM analysis and K-Means machine learning clustering, customers were segmented into four distinct groups based on purchasing behavior. The analysis also identifies which product categories drive the most revenue, which have the highest return rates, and what behavioral patterns predict customer churn.

Key Findings

- 1. Four distinct customer segments were identified.** K-Means clustering revealed: VIP customers (190 customers, 2.4% of base) averaging \$11,612 in spend; Loyal Mid-Tier (1,740 customers, 21.8%); At-Risk customers (1,289 customers, 16.1%) who haven't purchased in an average of 165 days; and New/Occasional buyers (4,781 customers, 59.8%) representing the largest conversion opportunity.
- 2. Electronics dominates revenue by a wide margin.** Electronics generated \$1.08M in total revenue — nearly 3x the next highest category (Clothing & Apparel at \$424K). Despite this, Electronics has a mid-range return rate of 8.5%, making it the strongest overall performing category.
- 3. Travel & Luggage and Pet Supplies carry the highest return rates.** At 10.0% and 9.9% respectively, these categories return nearly 1 in 10 orders — directly eroding margin. Clothing & Apparel had the lowest return rate at 7.7% despite being the second highest revenue category.
- 4. Churn rate is 8.9% with clear behavioral signals.** Churned customers averaged 94.9 days since their last purchase compared to 56.1 days for active customers — a 69% longer gap. Churned customers also spent 23.5% less on average (\$1,218 vs \$1,592), suggesting low engagement from the start is the strongest predictor of churn.

Business Recommendations

- 1. Launch a VIP retention program immediately.** 190 customers averaging \$11,612 in spend represent disproportionate revenue. Losing even 10% of this segment would have significant financial impact. Dedicated account management or exclusive offers are justified at this spend level.

2. Create re-engagement campaigns for the 1,289 At-Risk customers. These customers averaged 165 days since their last purchase. A targeted email campaign with personalized offers sent before the 180-day mark could recover a meaningful portion of this segment.

3. Investigate return rates in Travel & Luggage and Pet Supplies. At 10% and 9.9% respectively, these categories have structural return rate issues. Product quality, description accuracy, or sizing guides should be reviewed to reduce return volume.

4. Focus acquisition spend on converting New/Occasional buyers. At 59.8% of the customer base, even a 10% conversion to Loyal Mid-Tier would meaningfully grow recurring revenue. A post-first-purchase nurture sequence is the recommended approach.

Analysis conducted using Python (pandas, scikit-learn, matplotlib) and Tableau. Dataset: E-Commerce Customer Behavior & Sales 2020–2026 — Kaggle.